Panrui Ni (倪盼睿)

Email: panruini@gmail.com **Phone:** +86 13851650066
Website: panrui-ni.github.io **ORCID:** 0000-0003-0710-6994 **Google Scholar:** Profile
Current position: JSPS Postdoctoral Fellow (Standard), Waseda University, Japan

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Education

- **Ph.D. in Applied Mathematics** (Sep 2018 – Jun 2023)
  Shanghai Center for Mathematical Sciences, Fudan University, China.
  Dissertation: "Viscosity Solutions of Contact Hamilton–Jacobi Equations".
  Advisor: Prof. Jun Yan (严军).
- **B.Eng. in Engineering Mechanics** (Sep 2014 – Jun 2018)
  School of Civil Engineering, Southeast University, China.

Research Interests

My research focuses on nonlinear partial differential equations, particularly Hamilton–Jacobi equations, weak KAM theory, Aubry–Mather theory, and homogenization theory.

My research also involves contact Hamiltonian systems, mean field games, front propagation problems, and applications of Hamilton–Jacobi methods to dislocation dynamics.

Research & Work Experience

- JSPS Postdoctoral Fellowship (Standard) (Apr 2026 – present), Department of Mathematics, Faculty of Science and Engineering, Waseda University, Japan.
  Mentor: Prof. Hiroyoshi Mitake.
- Postdoctoral Researcher (Oct 2024 – Mar 2026), Graduate School of Mathematical Sciences, The University of Tokyo, Japan.
  Mentor: Prof. Hiroyoshi Mitake.
- Postdoctoral Researcher (Oct 2023 – Sep 2024), Institut de Mathématiques de Jussieu–Paris Rive Gauche, Sorbonne Université & Université Paris Cité, CNRS, France.
  Mentor: Prof. Maxime Zavidovique.

Publications **(* corresponding author)**

[1] **P. Ni***, *Quantitative homogenization of first-order ODEs*, Nonlinearity, 39 (2026), 105015.

[2] H. Mitake, **P. Ni***, *Quantitative homogenization of convex Hamilton–Jacobi equations with Neumann type boundary conditions*, Calculus of Variations and Partial Differential Equations, 65:154 (2026).

[3] H. Mitake, **P. Ni***, *Rate of convergence for homogenization of nonlinear weakly coupled Hamilton–Jacobi systems*, Journal of Differential Equations, 440:113442 (2025).

[4] **P. Ni**, M. Zavidovique*, *Nonlinear and degenerate discounted approximation in discrete weak KAM theory*, Mathematische Zeitschrift, 310:63 (2025).

[5] **P. Ni**, L. Wang*, J. Yan, *A representation formula of the viscosity solution of the contact Hamilton–Jacobi equation and its applications*, Chinese Annals of Mathematics, Series B, 46:449–480 (2025).

[6] **P. Ni**, L. Wang*, *A nonlinear semigroup approach to Hamilton–Jacobi equations–revisited*, Journal of Differential Equations, 403:272–307 (2024).

[7] **P. Ni**, L. Wang*, *Aubry–Mather theory for contact Hamiltonian systems III*, Science China Mathematics, 67:2541–2570 (2024).

[8] **P. Ni***, *Time periodic solutions of first order mean field games from the perspective of Mather theory*, Journal of Differential Equations, 412:881–901 (2024).

[9] **P. Ni***, *Weakly coupled Hamilton–Jacobi systems without monotonicity condition: A first step*, Communications on Pure and Applied Analysis, 23(7):961–983 (2024).

[10] **P. Ni**, K. Wang, J. Yan*, *Viscosity solutions of contact Hamilton–Jacobi equations with Hamiltonians depending periodically on unknown functions*, Communications on Pure and Applied Analysis, 22(2):668–685 (2023).

[11] **P. Ni***, *Multiple asymptotic behaviors of solutions in the generalized vanishing discount problem*, Proceedings of the American Mathematical Society, 151(12):5239–5250 (2023).

[12] **P. Ni**, K. Wang*, J. Yan, *A weakly coupled mean field games model of first order for k groups of major players*, Proceedings of the American Mathematical Society, accepted for publication.

[13] **P. Ni**, B. Shen*, *On variation of action integral in Finsler gravity*, Annals of Physics, 404(1):93–114 (2019).

Preprints

[1] **P. Ni***, *Sharp convergence rates for the vanishing discount problem with hyperbolic Aubry sets*, arXiv:2609.02779.

[2] H. Mitake, **P. Ni***, H. V. Tran, *Quantitative homogenization of convex Hamilton–Jacobi equations with u/ε-periodic Hamiltonians*, arXiv:2507.00663.

[3] H. Mitake*, **P. Ni**, H. V. Tran, *Vanishing discount and viscosity selection problems for mechanical Hamilton–Jacobi equations*, arXiv:2609.24009.

[4] A. Davini, **P. Ni***, J. Yan, M. Zavidovique, *Convergence/divergence phenomena in the vanishing discount limit of Hamilton–Jacobi equations*, arXiv:2411.13780.

[5] **P. Ni***, J. Yan, M. Zavidovique, *Selection of critical solutions in non-monotone vanishing discount problems*, arXiv:2609.24032.

[6] **P. Ni***, J. Yan, *A PDE formulation of Lyapunov stability for contact-type Hamilton–Jacobi equations*, arXiv:2604.24329.

[7] H. Mitake, **P. Ni***, *Optimal semiconcavity with fractional modulus for Hamilton–Jacobi equations with Neumann boundary conditions*, arXiv:2605.23248.

Fellowships & Research Projects

- JSPS Postdoctoral Fellowship (Standard), Japan Society for the Promotion of Science (JSPS), Apr 2026 – Mar 2027.
- National Natural Science Foundation of China (NSFC) General Program, "Global Dynamics of Contact Hamiltonian Systems without Monotonicity", Key Participant, Jan 2026 – Dec 2029.
- National Natural Science Foundation of China (NSFC) General Program, "Contact Hamiltonian Systems and Non-equilibrium Thermodynamics", Participant, Jan 2022 – Dec 2025.

Awards & Honors

- Outstanding Graduate of Shanghai, 2023.
- National Scholarship for Master's Students, Fudan University, 2019.
- Outstanding Student Award, Fudan University, 2019.

Invited Talks and Conference Presentations

- "Quantitative homogenization of first-order ODEs", Homogenization in PDE and Stochastic Processes, RIMS, Kyoto University, Kyoto, Japan (Aug 2026).
- "From dislocation dynamics to H–J equations: Homogenization and Orowan mechanism", Workshop on Recent Advances in Hamilton–Jacobi Equations, Shanghai Jiao Tong University, Shanghai, China (Jul 2026).
- "Quantitative Homogenization of Convex Hamilton–Jacobi Equations with u/ε-Periodic Hamiltonians", Viscosity Solutions of Differential Equations and Related Topics, RIMS, Kyoto University, Kyoto, Japan (Nov 2025).
- "Convergence/Divergence Phenomena in the Vanishing Discount Limit of Hamilton–Jacobi Equations", Recent Progress in Hamilton–Jacobi Equations and Related Topics, Nanjing University, Nanjing, China (Jun 2025).
- "On Discrete Nonlinear Vanishing Discount Problem", ANR Meeting, École Normale Supérieure de Lyon, Lyon, France (Jan 2024).
- "A Nonlinear Semigroup Approach to a Class of Nonmonotone Hamilton–Jacobi Equations", Conference on Differential Equations and Dynamical Systems, Beijing Institute of Technology, Beijing, China (Jul 2022).

Teaching Experience

- Teaching Assistant, Classical Mechanics (Spring 2019, Fall 2020), Fudan University, China.